

The Boundary Between Matter and Radiation

An Unresolved Classificatory Inconsistency in Modern Physics

Juliet Zhong

Independent Researcher, London, United Kingdom

ORCID: 0009-0006-5099-3671

Abstract

Modern physics employs two classificatory frameworks that have never been formally reconciled: the Standard Model's fermion/boson taxonomy and the inherited matter/radiation terminology of nineteenth-century physics. This paper argues that their coexistence conceals a genuine classificatory inconsistency. The term "radiation" includes electrons, classified as fermion matter particles, and photons, which are massless gauge bosons. Beta radiation is an electron flux; gamma radiation is a photon flux. Three arguments are developed. First, the historical formation of this inconsistency is traced: radiation terminology was extended to material and non-material entities without conceptual revision. Second, electron-photon interconversion through pair production and annihilation provides dynamical evidence against a categorical opposition between the two. Third, radiation is reinterpreted as a category of transferable field excitations rather than as an ontological opposite of matter. Within this framework, the electron is defined as a localised quantum excitation of the electron field, not itself a material structure but a constituent excitation from which material structures are formed. The ontological status of invariant mass remains an open foundational question. Matter is reassigned to the structural level, where the concept remains consistent. On this basis, the hypothesis is proposed that the electron's non-zero invariant mass may represent the effective behaviour of a more fundamental massless field structure in a bound configuration; three falsification conditions are stated.

Keywords: matter–radiation distinction; electron; photon; quantum field excitation; ontological classification; invariant mass; localised field configuration; beta radiation; pair production and annihilation; foundations of physics

1. Introduction: Two Coexisting Classification Systems

Every working physicist uses the words "matter" and "radiation." These terms appear in textbooks at every level, in the names of experimental programmes, in the framing of theoretical questions, and in the language with which physics communicates with adjacent sciences and with the public. Yet if one asks where, in the modern literature, the boundary between matter and radiation was precisely defined—defined in a way that is compatible with the Standard Model, consistent across subfields, and not merely inherited from nineteenth-century convention—the answer is surprisingly elusive.

The Standard Model of particle physics provides a rigorous and experimentally validated taxonomy of fundamental constituents. All known elementary particles are either fermions or bosons. Fermions—quarks and leptons—have half-integer spin and obey Fermi–Dirac statistics, which, through the spin–statistics connection, leads to the Pauli exclusion principle for identical fermions. Bosons—the gauge bosons of the strong, weak, and electromagnetic forces, and the Higgs scalar—have integer spin and obey Bose–Einstein statistics. The photon is a gauge boson of the electromagnetic interaction. The electron is a lepton, hence a fermion. This classification is rooted in the representation theory of the Lorentz group and has direct, measurable experimental content [Weinberg, 1995; Griffiths, 2008].

Alongside this taxonomy, a second and older classification continues in use: matter versus radiation. This distinction predates quantum mechanics and derives its intuitive force from classical mechanics and classical electrodynamics. Matter was the substance of atoms and molecules—localised, massive, and subject to Newtonian inertia. Radiation was the propagation of energy through the electromagnetic field—continuous, massless, and immaterial. The distinction felt absolute because, within classical physics, it was: no material object was also a wave in the electromagnetic field, and no electromagnetic wave had mass or occupied a definite position.

Quantum mechanics and the subsequent development of quantum field theory did not simply complicate this picture; they rendered the classical dichotomy untenable in several respects. Wave–particle duality showed that entities previously classified as material exhibited wave-like interference phenomena, while entities previously classified as radiation exhibited particle-like localisation and momentum. The field-theoretic framework of the Standard Model placed both categories within a common field-theoretic description: both the electron and the photon are described as excitations of quantum fields, differing in the type of field, the nature of the excitation, and the associated quantum numbers, but sharing the same fundamental mode of description.

What has not happened, in the century since these developments, is a formal revision of the matter/radiation terminology to reflect the new physics. The Standard Model provides a precise taxonomy, but physics education and scientific discourse continue to use the classical vocabulary alongside it, without reconciliation. The present paper argues that this unreconciled coexistence has produced a genuine terminological and classificatory inconsistency within the existing body of physics literature, and that this inconsistency has consequences for how fundamental questions about the nature of the electron and the photon are posed and answered.

The paper proceeds as follows. Section 2 traces the historical formation of the concept of radiation and documents how the term was extended to cover ontologically heterogeneous entities. Section 3 states the inconsistency precisely, using only definitions accepted in the standard literature. Section 4 examines electron–photon interconversion as a dynamical argument against the categorical opposition of the two. Section 5 proposes a reinterpreted classificatory basis. Section 6 provides a positive definition of the electron within this framework and addresses the question of matter at the constituent level. Section 7 addresses the status of matter at the structural level. Section 8 concludes.

Two terminological clarifications at the outset. "Radiation" in what follows refers to the term as used across physics broadly—electromagnetic radiation, radioactive emission, and related usages—not exclusively to ionising radiation or to cosmic rays. "Matter" refers to the classical category of massive, substantial entities, not to any cosmological technical usage. The inconsistency addressed is a problem of physical

ontology and classificatory language, not of experimental results: no measurement is contested.

2. How "Radiation" Acquired Its Meaning: A Historical Survey

To understand how the term "radiation" came to carry incompatible ontological commitments, it is necessary to trace its conceptual history from classical electrodynamics through the early twentieth century. The story is not one of confusion introduced by careless terminology; it is a story of terminology extended under experimental pressure, faster than conceptual revision could follow, and never subsequently brought into order.

In the framework of classical electrodynamics, as systematised by Maxwell, radiation meant electromagnetic waves [Maxwell, 1865]. These were disturbances of the electromagnetic field propagating through space at the speed of light, carrying energy and momentum but no rest mass. The characterisation was clear and its ontological commitment equally clear: radiation was immaterial. Matter, in the same framework, was the substance of atoms and molecules—subject to inertia, gravity, and contact forces. Radiation and matter interacted, but they were ontologically distinct.

This classical picture was under internal pressure before the discoveries of radioactivity. Planck's analysis of blackbody radiation (1900) required that electromagnetic energy be exchanged in discrete units [Planck, 1900]. Einstein's 1905 analysis of the photoelectric effect argued that the behaviour of light in that context was most economically explained by treating radiation as composed of discrete energy packets [Einstein, 1905]. The photon concept introduced a particle character to radiation. On the other side, Hertz's experimental confirmation of electromagnetic waves in 1888 had consolidated the continuous field picture [Hertz, 1888]. The ontological status of radiation was already undergoing revision before radioactivity forced the terminology further outward.

The discovery of radioactivity opened a second and decisive front of terminological pressure. Becquerel observed in 1896 that uranium salts spontaneously emitted something capable of producing detectable effects even when shielded from visible light [Becquerel, 1896]. The Curies systematically investigated this phenomenon and identified radium and polonium as intensely radioactive elements [Curie and Curie,

1898]. The emissions from radioactive materials were called "radiation" by analogy with electromagnetic radiation: they were invisible, penetrating, and detectable by their ionising effects. The naming was operationally motivated and experimentally convenient. It was not conceptually examined.

The differentiation of radioactive emissions into distinct types came rapidly. Rutherford showed in 1899 that uranium radiation contained at least two components with very different penetrating powers, labelled alpha and beta [Rutherford, 1899]. Villard identified a third component in 1900, eventually named gamma radiation [Villard, 1900]. The physical identification of these components took further years. Alpha particles were shown to be helium-4 nuclei—two protons and two neutrons, a composite massive object [Rutherford and Geiger, 1908]. Beta radiation was established as an electron flux: the same particles identified by Thomson in cathode ray experiments [Thomson, 1897; Chadwick, 1914]. Gamma radiation was established as high-energy electromagnetic radiation—photons—of the same fundamental nature as visible light and X-rays, differing only in frequency.

The significance of this identification for the matter/radiation distinction was not drawn at the time, and it has not been drawn since. Under the Standard Model, the three "radiations" of radioactive decay represent three entirely different ontological categories. Alpha radiation is a composite nuclear bound state—matter in the fullest classical sense. Beta radiation is an electron: a fundamental fermion, conventionally classified as a matter particle in Standard Model discussions. Gamma radiation is a photon: a gauge boson, massless, not classified as a matter particle in the Standard Model. The word "radiation" had been extended, by experimental necessity and conceptual inertia, to cover all three—a helium nucleus, an electron, and a photon—without any acknowledgement that this grouping placed ontologically heterogeneous entities under a single term.

The terminology was actively reinforced by the institutional and pedagogical structure of early twentieth-century physics. Nuclear physics and radiation physics developed as applied fields with strong connections to medicine, engineering, and military applications. In these contexts, the operational definition of "radiation"—something that penetrates matter, ionises gases, and can be detected by its effects—was entirely adequate. The operational aims of these fields did not require a resolution of their

deeper ontological classification. The Standard Model, when it arrived, provided the right conceptual tools, but it was developed in a different scientific context, and no formal revision of the broader radiation vocabulary followed.

The 1927 Solvay Conference illustrates how acute the terminological tension had become by the mid-1920s. The central debates concerned the interpretation of quantum mechanics—the status of the wave function, the relationship between wave and particle descriptions [Bacciagaluppi and Valentini, 2009]. The wave–particle duality of both matter and radiation was acknowledged, but it was treated as a puzzle about the relationship between descriptions, not as a signal that the matter/radiation distinction itself required revision. The language of "radiant energy" for electromagnetic waves and "corpuscular rays" for particle beams persisted as separate traditions, both in tension with a unified quantum field picture that was only beginning to be articulated.

The conclusion of this historical survey is as follows. The term "radiation" in modern physics carries the trace of two distinct conceptual traditions: the electromagnetic tradition, in which radiation is immaterial wave propagation, and the nuclear/radiological tradition, in which radiation designates any penetrating emission from a radioactive source, regardless of material character. These two traditions were never reconciled. The result is a term that simultaneously includes and excludes material particles depending on which tradition is operative. This is the historical root of the inconsistency; the next section states it with logical precision.

3. The Inconsistency Made Explicit: Matter Inside Radiation

The historical survey of the preceding section documents how the inconsistency arose. The present section states it precisely, using only propositions individually accepted in the standard literature. No new theoretical claim is introduced; the classificatory inconsistency is entirely internal to the existing terminology of physics.

Consider the following four propositions, each endorsed by mainstream physics texts.

Proposition (a): In conventional Standard Model terminology, electrons belong to the fermionic matter sector. Electrons are members of the first generation of leptons, with non-zero rest mass ($0.511 \text{ MeV}/c^2$), spin- $1/2$, and charge $-e$. Leptons, together with quarks, constitute the fermionic sector conventionally associated with matter particles,

as opposed to the gauge bosons, which are force carriers [Griffiths, 2008; Halzen and Martin, 1984; Particle Data Group, 2022]. This classification is grounded in the mathematical structure of the theory and in the physical differences between fermions and bosons.

Proposition (b): In nuclear physics and radiation physics, beta radiation is defined as the emission of electrons (beta-minus) or positrons (beta-plus) from unstable atomic nuclei. Beta radiation is universally listed as one of the three principal types of ionising radiation, alongside alpha and gamma [Krane, 1988; Knoll, 2010; Turner, 2007]. Health physics, radiation protection standards, and nuclear engineering texts define beta radiation as an electron flux.

Proposition (c): From (a) and (b) together, the electron is simultaneously a matter particle (Standard Model) and a component of radiation (nuclear physics). The same physical entity is included within a category of radiation in one disciplinary context while classified as matter in another, without a corresponding terminological reconciliation in either body of literature.

Proposition (d): The photon is classified as radiation—electromagnetic radiation in the general case, and gamma radiation in the high-energy case. The photon is a gauge boson of the electromagnetic interaction, with zero rest mass, spin-1, and no electric charge. It is not classified as a matter particle in the Standard Model [Weinberg, 1995; Particle Data Group, 2022].

The logical upshot is direct. If matter and radiation are mutually exclusive categories, then proposition (c) is a contradiction: the electron cannot be both a matter particle and a component of radiation. If they are not mutually exclusive—if something can be both matter and radiation—then the matter/radiation boundary is not the dichotomy it purports to be, and it requires formal redefinition. Either conclusion demonstrates that the existing classificatory framework is inadequate.

Two objections present themselves. The first is that "radiation" in the nuclear physics sense and "radiation" in the electromagnetic sense are simply two different technical usages of the same word—homonymy rather than inconsistency. This objection, if accepted, concedes the central point: the same scientific vocabulary carries two incompatible technical meanings in different subfields without formal disambiguation.

Standard texts in nuclear physics define radiation as inclusive of both electrons and photons without distinguishing two separate senses, treating the category as unified while assigning it members that belong to ontologically different Standard Model classes. This objection shifts the issue from contradiction to unresolved terminology; the classificatory ambiguity remains, and with it the consequence: the matter/radiation distinction does not provide a coherent basis for fundamental classification.

The second objection is that the matter/radiation distinction was never intended as a fundamental ontological boundary—only as a practical operational classification. This objection is historically questionable; the distinction has been treated as ontologically significant throughout its history and in foundational discussions. But even if it were accepted, it would deepen rather than resolve the problem: a distinction that is merely operational cannot legitimately bear the ontological weight that has been placed on it. The task of the remainder of this paper is to provide a replacement classification that does not carry this internal inconsistency.

There is a further, less visible consequence worth noting. The partial coincidence between the Standard Model's fermion/boson distinction and the classical matter/radiation distinction has allowed the inconsistency to persist unnoticed. In standard cases, fermions are material (quarks, leptons) and bosons are radiation-like (photons, gauge bosons), and the electron fits this casual mapping. But the mapping is not identity. The fermion/boson distinction is a classification of statistical behaviour and spin; the matter/radiation distinction is an ontological claim about the nature of physical entities. Treating them as interchangeable has obscured the fact that "radiation" in nuclear physics is doing something ontologically different from "radiation" in electrodynamics. The solution is not to continue conflating them but to adopt a classificatory basis that generates no inconsistency—which is the task of Section 5.

4. Conditional Reversibility: Electron–Photon Interconversion

The terminological argument of the preceding section establishes that the matter/radiation boundary is internally inconsistent within existing physics, using only standard definitions. The present section introduces a second, dynamical argument: the known physics of electron–photon interconversion provides additional grounds for

questioning whether the electron and the photon should be placed in categorically opposed ontological classes.

4.1 Electron–Positron Annihilation

When an electron and a positron meet with sufficiently low relative velocity, they may annihilate:

$$e^{-} + e^{+} \rightarrow 2\gamma$$

The products are two photons. The two-photon requirement is not arbitrary; it follows from the simultaneous conservation of energy, linear momentum, and angular momentum. In the centre-of-momentum frame, a single photon would need to carry zero total momentum while carrying the full energy of the annihilating pair—a condition that is kinematically impossible for a massless particle. Two photons emitted in opposite directions with equal energies satisfy all conservation laws [Dirac, 1930; Berestetskii, Lifshitz, and Pitaevskii, 1982]. Each photon carries approximately 0.511 MeV in the case where the electron and positron are initially at rest, corresponding exactly to the electron rest-mass energy. The amplitude contains two QED vertices and is therefore of order e^2 . It has been verified to great precision and forms the physical basis of positron emission tomography [Knoll, 2010].

What this process demonstrates is that the entire energy and all the quantum numbers of an electron–positron pair can be transferred completely into photons. The electron–positron state conversion is total under the allowed conditions: nothing of the pair's quantum state, beyond what is encoded in the two outgoing photons, persists in the final state.

4.2 Pair Production

The reverse process exists:

$$\gamma \rightarrow e^{-} + e^{+}$$

A high-energy photon can produce an electron–positron pair. This process requires a condition that the simple notation does not display. A single photon in vacuum cannot produce a pair while conserving both energy and momentum simultaneously: the photon has zero invariant mass while the pair has invariant mass of at least $2m_e c^2 =$

1.022 MeV. The momentum mismatch must be absorbed by an external agent, typically the Coulomb field of a nucleus, or alternatively by a second photon:

$$\gamma\gamma \rightarrow e^- + e^+$$

where the combined four-momentum of the two photons can carry sufficient invariant mass [Berestetskii, Lifshitz, and Pitaevskii, 1982; Jauch and Rohrlich, 1955]. Both processes are second-order in the QED perturbation expansion.

The conditionality of pair production is important and must not be overstated. Electron–photon interconversion is not a free or spontaneous process; it is a conditional, conservation-law-governed quantum transition. Overstating the reversibility would misrepresent the physics. The present paper does not claim that electrons and photons are identical entities, or that any photon can spontaneously become an electron.

4.3 The Classificatory Implication

What the conditionality does not alter is the classificatory significance of the interconversion. The fact that an electron–positron pair can be completely and reversibly converted into photons—and that photons under specified conditions produce electron–positron pairs—means that the two kinds of entity are dynamically coupled at the most fundamental level. They are connected by actual physical processes in which one form of field excitation becomes another.

Physics has repeatedly found that when two apparently distinct entities can be mutually converted under specifiable conditions, a deeper theoretical unification follows. In some historical cases, conditional interconvertibility between apparently distinct entities has motivated the search for a deeper classificatory framework. The pattern is not limited to gauge unification. The proton and neutron, once treated as distinct fundamental entities, were recognised as two isospin states of a common nucleon structure, a revision that preceded and motivated the quark model [Heisenberg, 1932]. The recognition that the photon and the electroweak gauge bosons are different manifestations of a common gauge structure is one such case [Glashow, 1961; Weinberg, 1967; Salam, 1968]. The analogy with the electron–photon case is not exact—annihilation and pair production involve particle creation, not rotation within a multiplet—but the broader pattern provides motivation: interconvertibility has in several instances preceded classificatory revision.

The argument of this section is bounded precisely. The existence of conditional reversible interconversion between electrons and photons provides a physical reason to question their placement in categorically opposed ontological classes. A classification system in which entities are placed in mutually exclusive fundamental categories, but in which one can become the other under specifiable conditions, is a classification system under tension. That tension is what the proposal in Section 5 is designed to resolve.

5. A Reinterpreted Category: Field Excitation as the Basis of Classification

The preceding sections have established two lines of argument against the classical matter/radiation dichotomy. The terminological argument shows that the same physical object—the electron—is simultaneously classified as matter and as radiation within the existing literature. The dynamical argument shows that entities placed in opposed ontological categories undergo conditional reversible interconversion, a pattern that has historically motivated classificatory unification. The present section proposes a replacement classificatory basis.

5.1 The Proposed Reinterpretation

The proposal is as follows. Radiation is reinterpreted here as a category of quantum excitations capable of carrying energy, momentum, and conserved quantum numbers across macroscopic physical boundaries, rather than as an ontological opposite of matter. This reinterpretation does not claim that every quantum field excitation is radiation in any prior sense of the word; it claims that the word "radiation" should no longer be defined as the ontological negation of matter, and should instead be understood as covering the class of field excitations that can propagate, transfer, or be exchanged across space. The reinterpretation targets the matter/radiation boundary, not the Standard Model taxonomy of fields.

Under this reinterpretation, radiation admits multiple modes, of which the following are empirically known and are directly relevant to the argument of this paper. The free-propagating mode is exemplified by the photon: a quantum excitation of the electromagnetic field that propagates through space without restriction, encompassing the full spectrum from radio waves through microwave, infrared, visible, ultraviolet, X-ray, and gamma-ray frequencies. All of these are photons of different energies, not

different ontological types. The localised mode is exemplified by the electron: a quantum excitation of the electron field that can exist in localised wave-packet states and is characterised by associated quantum numbers. The reinterpretation does not exclude other forms of quantum field excitation within this broad category; the neutrino, for instance, is a propagating fermionic excitation with near-zero invariant mass, and its classificatory status under the present framework is a question left for further analysis. Observable differences between these modes—in invariant mass, spin, electric charge—are properties of the modes; they are not ontological boundaries that place the modes in opposed fundamental categories.

The reinterpretation resolves the terminological inconsistency documented in Section 3. Under the new reading, beta radiation and gamma radiation are both radiation in a coherent and unified sense: one is a flux of localised field excitations (electrons), the other a flux of free-propagating field excitations (photons). They are different modes of field excitation, not different kinds of radiation that happen to share a name by historical accident. Alpha radiation—a helium-4 nucleus, a composite bound state—is radiation in the operational sense of a penetrating emission, but is not a fundamental field excitation in the same sense; its status is that of a composite structure, which is addressed in Section 7. The inconsistency arose because "radiation" was extended to composite structures and fundamental excitations alike, without distinguishing the two levels.

5.2 Scope and Limits of the Reinterpretation

The reinterpretation requires careful demarcation. It does not assert that all quantum field excitations are radiation. Quarks, for instance, do not propagate freely; they are permanently confined within hadrons by the strong interaction, and it would be misleading to describe a confined quark as "radiation" in any standard sense. The Higgs field is responsible for electroweak symmetry breaking and fermion mass generation; the Higgs boson is its quantum excitation and does not propagate in the same sense as the photon or the free electron. The massive gauge bosons W and Z are short-range force carriers. The reinterpretation focuses specifically on excitations that are transferable or propagating in the relevant physical sense—those that can travel macroscopic distances and carry energy, momentum, and quantum numbers across space. This scope is broad enough to include electrons and photons, which are the

entities at issue, without committing to a universal equation of "field excitation" with "radiation."

The reinterpretation also does not claim that the Standard Model's fermion/boson distinction is incorrect or should be replaced. The distinction between fermionic and bosonic excitations is real, important, and experimentally grounded: it governs statistical behaviour, the Pauli exclusion principle, and the population of quantum states. What the reinterpretation denies is that the fermion/boson distinction coincides with the matter/radiation distinction in a way that makes the latter a fundamental ontological category. The electron is a fermion; that is a correct and useful classification. The claim that being a fermion places the electron on the matter side of a matter/radiation ontological boundary is what is challenged here.

5.3 Compatibility with the Higgs Mechanism

The Higgs mechanism provides the Standard Model's account of why the electron has a non-zero invariant mass while the photon does not. The electromagnetic U(1) gauge symmetry requires the photon to be massless; a mass term would break local gauge invariance [Weinberg, 1995; Higgs, 1964; Englert and Brout, 1964]. The electron acquires its mass through the Yukawa coupling to the Higgs field: the coupling term $-\gamma_e \bar{e}_L \Phi e_R + \text{h.c.}$ produces a mass term when the Higgs field acquires a non-zero vacuum expectation value through spontaneous symmetry breaking. This mechanism is well established and is not questioned here.

The Higgs mechanism is compatible with the proposed reinterpretation. It specifies how the electron excitation mode acquires its characteristic mass parameter; it does not specify, in an ontological sense, whether the resulting mass is an irreducible intrinsic property of the electron field or an effective parameter of a particular excitation configuration. The Higgs mechanism provides the Standard Model explanation of the electron mass parameter; whether additional ontological interpretation is required is a separate foundational question. The reinterpretation proposed here does not conflict with the Higgs mechanism; it situates that mechanism within a broader classificatory account and leaves that foundational question open. That question is addressed in the following section.

6. The Electron: Constituent Excitation and Its Classification

The reinterpretation of Section 5 places the electron and the photon within the same broad classificatory territory—both are transferable or propagating field excitations—while distinguishing them by mode and associated quantum numbers. This section fulfils the obligation that placement creates: it gives the electron a positive definition within the new framework. The central argument of this section concerns a principle that is prior to any discussion of the electron's quantum numbers or mass: the classification of a constituent cannot be inferred from the classification of the composite system it participates in forming. This principle, when applied to the electron, dissolves the most persistent source of confusion about its ontological status.

6.1 Constituent versus Composite: The Primary Distinction

The electron participates in forming atoms. In the hydrogen atom, the electron occupies the bound states of the Coulomb potential defined by the proton; in heavier atoms, electrons fill the shells of the nuclear potential, and their configuration determines the atom's chemical identity. From atoms, electrons participate in forming molecules through covalent, ionic, and metallic bonding. From molecules, all the composite structures of condensed matter physics, chemistry, biology, and engineering are built. The electron is, in this sense, a constituent of what we ordinarily call material reality.

The question is whether this constitutive role determines the electron's own classification. The answer is that it does not, and cannot. The classification of a component is not determined by the classification of the composite structure that the component helps to form. This is a general principle of classificatory logic, not a claim specific to physics: the category to which a part belongs is established at the level of parts, not at the level of wholes. A word is not itself a sentence because sentences are composed of words; a cell is not itself an organism because organisms are composed of cells. Applying the classification of the composite to its constituents is a category error—the confusion of two levels of description that must be kept distinct.

Applied to the electron: the fact that electrons participate in forming atoms, and that atoms constitute what we call matter, does not make the electron itself a material structure in the relevant sense. The electron is a constituent excitation from which material structures are formed; it is not itself a material structure. The two levels are distinct. The ontological classification of a constituent is not determined by the composite structure it participates in forming. This is the distinction between the

fundamental excitation level and the composite structure level. At the level of fundamental excitation, the electron is a localised quantum excitation of the electron field. At the level of composite structure, atoms and molecules formed from such excitations constitute matter. Conflating these two levels is precisely what the classical matter/radiation framework has done, by classifying the electron as a matter particle on the basis of its role in composite matter, and then treating that classification as if it established the electron's fundamental ontological kind.

This distinction resolves the apparent paradox that motivates much of the difficulty. It seems obvious that electrons are material: they are what atoms are made of, and atoms are material. But the appearance of paradox dissolves when the levels are separated. Atoms are material because they are stable composite structures with mass, volume, and chemical identity. The electron is a constituent of those structures. Its being a constituent does not transfer the classification of the composite to the constituent; it means that the composite is built from something that is, at a more fundamental level, a field excitation. The electron is not less real or less physical for being classified as a field excitation rather than as matter; it is classified at the level at which the classification is appropriate.

6.2 Positive Definition of the Electron

Within the framework proposed in this paper, the electron is a localised quantum excitation of the electron field. This is its positive definition, stated at the level of fundamental excitation rather than at the level of composite structure.

"Localised" specifies the excitation mode: the excitation energy is confined to a region of space, in contrast to the free-propagating mode of the photon, in which the excitation propagates indefinitely through space. "Quantum excitation of the electron field" situates the electron in the framework of quantum field theory: it is not a classical point particle, nor a classical wave, but a discrete excitation of a fundamental field.

The observable characteristics of the electron excitation mode are: invariant mass $m_e = 0.511 \text{ MeV}/c^2$; spin quantum number $s = 1/2$, making it a fermion subject to the Pauli exclusion principle; and electric charge $q = -e$, where $e = 1.602 \times 10^{-19} \text{ C}$. The positron—the electron's antiparticle—is the antiparticle excitation of the same field: identical mass and spin, charge $+e$. For comparison: the photon is a free-propagating

quantum excitation of the electromagnetic field, with zero invariant mass, spin-1, and zero electric charge.

These definitions are definitions of excitation modes within the same broad category. The electron and photon differ in the field to which they belong, in the mode of excitation (localised versus free-propagating), and in their associated quantum numbers. None of these differences constitutes an ontological boundary of the kind the classical matter/radiation distinction claimed to mark. They are differences among modes within a shared fundamental description.

6.3 The Invariant Mass: An Open Foundational Question

The most salient observable difference between the electron and the photon is the electron's non-zero invariant mass. The photon's zero mass is required by the U(1) gauge symmetry of electrodynamics: a mass term for the photon would violate local gauge invariance [Weinberg, 1995]. The electron's non-zero mass is generated by the Higgs mechanism through the Yukawa coupling, as described in Section 5.3. The Standard Model specifies the acquisition mechanism and fixes the numerical value of m_e .

What the Higgs mechanism does not settle is the ontological status of the mass it generates: whether the electron's invariant mass should be understood as an irreducible intrinsic property of the electron field, or as an effective parameter that characterises the localised excitation mode and could in principle be understood as arising from a more fundamental configuration. This is an open foundational question, distinct from the question of the mass's numerical value or its acquisition mechanism. The Higgs mechanism answers "how does the electron acquire this mass value" but not "what does it mean for a localised excitation mode to have mass rather than a propagating one."

If the electron and the photon are both field excitations in different modes, then the question of why one mode has non-zero invariant mass and the other does not is a question about the relationship between excitation mode and mass—a foundational question that the Standard Model describes quantitatively rather than resolves at the interpretive level. Whether the answer lies in a deeper field-theoretic structure, in the topology of excitation configurations, or in some other mechanism is not addressed here. The contribution of the present classification is to identify this as a well-formed

open question and, further, to motivate a specific hypothesis about the relationship between excitation mode and invariant mass.

The classification proposed in this paper makes this question more precisely formulable and motivates a specific hypothesis. The electron's localised excitation mode—characterised by non-zero invariant mass—may represent the effective behaviour of a more fundamental massless field structure in a bound configuration, rather than an irreducible intrinsic property of the electron field. This is proposed as a hypothesis, not as a conclusion of the classificatory argument above. The classificatory argument stands independently of whether this hypothesis is confirmed or refuted.

Three falsification conditions are stated:

First, if experiment confirms that the electron possesses an irreducible internal structural scale and a mass component not accounted for by Higgs coupling—a component that cannot be attributed to any bound configuration of a more fundamental structure—the hypothesis requires revision.

Second, if future theoretical work demonstrates that no bound configuration of any massless field can produce half-integer spin and quantised electric charge simultaneously, the hypothesis is excluded.

Third, the hypothesis requires that the electron's invariant mass be strictly constant across all bound and free environments; any experimentally confirmed environment-dependence of the electron's invariant mass would falsify the specific form of the hypothesis as stated here.

This hypothesis is not a precondition of the classificatory proposal of this paper. It is a direction that the classification makes clearly formulable: if the electron and the photon are both field excitations in distinct modes, the question of whether the localised mode with invariant mass and the free-propagating massless mode share a deeper field-theoretic origin becomes a well-posed research question rather than a category error.

7. What Then Is "Matter"?

If the electron is classified as a constituent field excitation rather than as a matter particle in the classical sense, the concept of matter requires relocation. The relocation

proposed here is clarifying rather than eliminative: matter is assigned to the structural level, where it has always done its genuine explanatory work, and is removed from the level of fundamental excitations, where it has generated the inconsistency this paper has documented.

Consider what the concept of matter actually explains or delineates in scientific practice. When physicists, chemists, engineers, and biologists speak of matter, they are concerned with substances that have mass, occupy volume, possess chemical identity, and interact with other substances through forces that can be described by the laws of chemistry and condensed matter physics. Solid matter has a crystal structure; liquid matter has surface tension; gaseous matter obeys an equation of state; plasma is a gas of ionised constituents. These descriptions apply to systems composed of large numbers of atoms and molecules, interacting through electromagnetic forces at the atomic scale. The four states of matter—solid, liquid, gas, plasma—are descriptions of collective behaviour, statistical properties of many-particle systems, not intrinsic properties of the fundamental constituents.

The atom is the basic unit of this level of description. An atom is a bound state of a nucleus (itself a composite of quarks and gluons) and electrons bound to the nucleus by the electromagnetic interaction. From atoms, molecules are formed by chemical bonding. From molecules, all macroscopic structure is built. At every level of this hierarchy, the description is of composite structure—of how constituents are organised and interact—not of what the constituents fundamentally are.

Within the classificatory proposal of this paper, matter is treated primarily as a structural concept: it refers to composite, stable configurations of field excitations, particularly the configurations of electrons and quark-composite nuclei that together form atoms and the structures built from atoms. This definition preserves everything that the concept of matter needs to do in chemistry, condensed matter physics, biology, and engineering. It also eliminates the inconsistency: the electron, as a fundamental excitation, is not itself a material structure in the structural sense defined here; it is a constituent excitation from which material structures are formed.

This reassignment has a clarifying consequence for the four-state taxonomy. It has always been slightly anomalous to ask what "state of matter" a single electron occupies: the question applies the wrong tool at the wrong level. Solid, liquid, gas, and plasma

are descriptions of many-body collective behaviour; they apply to atoms and to systems of atoms, not to fundamental excitations. On the present account, this is transparent rather than puzzling: the four states apply at the structural level, where matter is defined, and do not reach to the excitation level, where the electron resides. A single electron occupies no state of matter for the same reason that a single constituent of a structure cannot itself instantiate the structure's macroscopic properties—the description does not apply at that level.

The concept of matter is preserved rather than abolished. What has changed is that matter is no longer treated as a fundamental ontological category co-extensive with fermions, or with "that which has mass," or with "that which is opposed to radiation." It is a macroscopic, structural, emergent category—the appropriate description of organised composite structures built from fundamental excitations. The classical intuition that matter is the substantial stuff of the world is preserved at the level where that intuition is correct: the level of atoms, molecules, and the structures composed of them. What is revised is the extension of that category downward to fundamental excitations, where it was never correctly applicable and where it produced the inconsistency documented in this paper.

8. Conclusion

This paper has pursued three arguments whose convergence supports a unified proposal.

The first argument is historical and terminological. The term "radiation" entered physics in two distinct traditions—the electromagnetic tradition, in which radiation is immaterial wave propagation, and the nuclear/radiological tradition, in which radiation designates any penetrating emission regardless of material character. When radioactivity was characterised, the word "radiation" was extended without conceptual revision to cover helium nuclei, electrons, and photons alike. The Standard Model subsequently provided a precise taxonomy distinguishing these three—composite nuclear state, fundamental fermion, massless gauge boson—but did not formally retire or correct the radiation vocabulary. The result has been more than a century during which the same scientific community classifies the electron as a matter particle in one context and as a component of radiation in another, without acknowledging the inconsistency.

The second argument is dynamical. Electron–positron annihilation and pair production demonstrate that the electron and the photon are not categorically incommensurable: under specifiable conditions, electron–positron states and photon states can undergo conditional reversible conversion. The conditionality of pair production—requiring an external field or a second photon—has been carefully noted. What the conditionality does not alter is the classificatory significance of the interconversion: the pattern of conditional reversible conversion between apparently distinct entities has, in the history of physics, repeatedly motivated and preceded classificatory unification.

The third argument is constructive. Radiation is reinterpreted as a category of quantum excitations capable of carrying energy, momentum, and conserved quantum numbers across macroscopic physical boundaries, rather than as an ontological opposite of matter. Within this reinterpretation, the electron is given a positive definition as a localised quantum excitation of the electron field—not itself a material structure, but a constituent excitation from which material structures are formed. The distinction between constituent and composite is the philosophical core of the proposal: the classification of a fundamental constituent need not be inherited from the classification of the composite structures that it forms. The concept of matter is accordingly reassigned to the structural level—composite stable configurations of field excitations—where the four-state taxonomy of solid, liquid, gas, and plasma finds its proper domain of application. The electron's non-zero invariant mass, and the question of why the localised excitation mode has mass while the free-propagating mode does not, is identified as an open foundational question that the present classification makes more precisely formulable.

The contribution is classificatory: it fills a gap that has persisted in physical ontology for more than a century, clarifies the relationship between two frameworks that have been coexisting without formal reconciliation, and states more exactly the question that follows. If the electron and the photon are both field excitations in distinct modes, the question of whether those modes arise from a single underlying field structure becomes clearly formulable. The classical matter/radiation framework, with its internal inconsistency and its ontological commitments inherited from the nineteenth century, has not permitted this question to be stated with the precision it deserves. The framework proposed here does.

Acknowledgements

The author used Claude (Anthropic), GPT (OpenAI), and Gemini (Google) to assist with manuscript drafting, literature synthesis, and language editing. All conceptual frameworks, logical arguments, and final conclusions were developed and verified by the author, who assumes full responsibility for the integrity of the work.

Declarations

Funding: This research received no external funding.

Conflicts of interest: The author declares no conflicts of interest.

Data availability: No new observational data were generated or analysed in this study. All referenced datasets are publicly available from the sources cited.

Author contributions: Juliet Zhong: conceptualisation, formal analysis, writing.

References

- [Bacciagaluppi and Valentini, 2009] Bacciagaluppi, G. and Valentini, A. (2009). Quantum Theory at the Crossroads: Reconsidering the 1927 Solvay Conference. Cambridge University Press, Cambridge.
- [Becquerel, 1896] Becquerel, H. (1896). Sur les radiations émises par phosphorescence. *Comptes Rendus*, 122:420–421.
- [Berestetskii, Lifshitz, and Pitaevskii, 1982] Berestetskii, V. B., Lifshitz, E. M., and Pitaevskii, L. P. (1982). Quantum Electrodynamics. 2nd ed. Pergamon Press, Oxford.
- [Chadwick, 1914] Chadwick, J. (1914). Intensitätsverteilung im magnetischen Spektrum der β -Strahlen von Radium B + C. *Verhandlungen der Deutschen Physikalischen Gesellschaft*, 16:383–391.
- [Curie and Curie, 1898] Curie, P. and Curie, M. (1898). Sur une substance nouvelle radio-active, contenue dans la pechblende. *Comptes Rendus*, 127:175–178.
- [Dirac, 1930] Dirac, P. A. M. (1930). A theory of electrons and protons. *Proceedings of the Royal Society A*, 126:360–365.
- [Einstein, 1905] Einstein, A. (1905). Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt. *Annalen der Physik*, 17:132–148.

- [Englert and Brout, 1964] Englert, F. and Brout, R. (1964). Broken symmetry and the mass of gauge vector mesons. *Physical Review Letters*, 13:321–323.
- [Glashow, 1961] Glashow, S. L. (1961). Partial-symmetries of weak interactions. *Nuclear Physics*, 22:579–588.
- [Griffiths, 2008] Griffiths, D. (2008). *Introduction to Elementary Particles*. 2nd ed. Wiley-VCH, Weinheim.
- [Halzen and Martin, 1984] Halzen, F. and Martin, A. D. (1984). *Quarks and Leptons: An Introductory Course in Modern Particle Physics*. Wiley, New York.
- [Heisenberg, 1932] Heisenberg, W. (1932). Über den Bau der Atomkerne. *Zeitschrift für Physik*, 77:1–11.
- [Hertz, 1888] Hertz, H. (1888). Über Strahlen elektrischer Kraft. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, pp. 1297–1307.
- [Higgs, 1964] Higgs, P. W. (1964). Broken symmetries and the masses of gauge bosons. *Physical Review Letters*, 13:508–509.
- [Jauch and Rohrlich, 1955] Jauch, J. M. and Rohrlich, F. (1955). *The Theory of Photons and Electrons*. Addison-Wesley, Cambridge, MA.
- [Knoll, 2010] Knoll, G. F. (2010). *Radiation Detection and Measurement*. 4th ed. Wiley, Hoboken.
- [Krane, 1988] Krane, K. S. (1988). *Introductory Nuclear Physics*. Wiley, New York.
- [Maxwell, 1865] Maxwell, J. C. (1865). A dynamical theory of the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 155:459–512.
- [Particle Data Group, 2022] Particle Data Group (2022). Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2022:083C01.
- [Planck, 1900] Planck, M. (1900). Zur Theorie des Gesetzes der Energieverteilung im Normalspektrum. *Verhandlungen der Deutschen Physikalischen Gesellschaft*, 2:237–245.
- [Rutherford, 1899] Rutherford, E. (1899). Uranium radiation and the electrical conduction produced by it. *Philosophical Magazine*, 47:109–163.
- [Rutherford and Geiger, 1908] Rutherford, E. and Geiger, H. (1908). An electrical method of counting the number of α particles from radio-active substances. *Proceedings of the Royal Society A*, 81:141–161.
- [Salam, 1968] Salam, A. (1968). Weak and electromagnetic interactions. In Svartholm, N. (ed.), *Elementary Particle Theory*, pp. 367–377. Almqvist and Wiksell, Stockholm.
- [Thomson, 1897] Thomson, J. J. (1897). Cathode rays. *Philosophical Magazine*, 44:293–316.
- [Turner, 2007] Turner, J. E. (2007). *Atoms, Radiation, and Radiation Protection*. 3rd ed. Wiley-VCH, Weinheim.

- [Villard, 1900] Villard, P. (1900). Sur la réflexion et la réfraction des rayons cathodiques et des rayons déviabiles du radium. *Comptes Rendus*, 130:1010–1012.
- [Weinberg, 1967] Weinberg, S. (1967). A model of leptons. *Physical Review Letters*, 19:1264–1266.
- [Weinberg, 1995] Weinberg, S. (1995). *The Quantum Theory of Fields*. Vol. 1. Cambridge University Press, Cambridge.